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Assume $t_B = t_1 + \theta$, round trip = $[t_2 - (t_1 + \theta)] + [t_4 - (t_3 - \theta)]$
 $= 1742789000 - 1741688100 + 1741688115 - 1742789005 = 10$

So, $t_2 - (t_1 + \theta) = 5 \Rightarrow \theta = 1742789000 - 1741688100 - 5$
 $\Rightarrow \theta = 1100895$

2. 1st round: F2, 2nd: F3, 3rd: F1

3. A, because if we take timestamp at C, it will have delay jitter caused by software stack (application/OS/MAC layer).

4. Transparent clock record the time spent in transmission through each network device, and the clock measure and adjust for the delay as packets pass through the clock.

6. $g(D)$ has degree of 4 $\Rightarrow S(D) \cdot D^4 = D^7 + D^5$

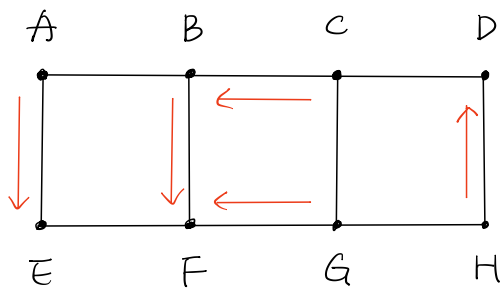
$$\begin{array}{r}
 D^3 + + + 1 \\
 D^4 + D^2 + D + 1 \overline{) D^7 + + D^5} \\
 \underline{D^7 + + D^5 + D^4 + D^3} \\
 D^4 + D^3 \\
 \underline{D^4 + + D^2 + D + 1} \\
 D^3 + D^2 + D + 1
 \end{array}$$

The remainder $c(D) = D^3 + D^2 + D + 1$, $x(D) = S(D) \cdot D^4 + c(D)$

$\Rightarrow x(D) = D^7 + D^5 + D^3 + D^2 + D + 1$

7. False. With wired-AND, need all clocks to be HIGH, then the synchronized clocks becomes HIGH.

8.



First, find receiver in these transmissions : B, D, E, F

Primary : only F, because B, D, E don't receive by multiple sender.

Secondary : B $\Rightarrow$ Yes, because A is the sender.

D $\Rightarrow$ Yes, because C is the sender.

E $\Rightarrow$ No, because no other sender in E's range.

So, $n_p = 1$, $n_s = 2$.

$$\begin{aligned}
 9. \quad & wt(00000000) = 0 \\
 & wt(0011101) = 4 \\
 & wt(0100111) = 4 \\
 & wt(0111010) = 4 \\
 & wt(1001110) = 4 \\
 & wt(1010011) = 4 \\
 & wt(1101001) = 4 \\
 & wt(1110100) = 4
 \end{aligned}$$

So the Hamming weight for the set of code words is $\min\{4, 4, 4, 4, 4, 4, 4\} = 4$.